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Course: biotechnology

Task: 1

Report on Crisper Cas9 Gene Editing in Cancer

Introduction

CRISPR-Cas9 is a revolutionary gene-editing technology that allows scientists to alter, delete, or replace specific sequences of DNA

within living organisms. Adapted from a natural bacterial immune system, it acts as a highly precise biological “find and replace” tool for the genome Supercharging T Cells: Scientists harvest a patient’s immune cells (T cells) and use CRISPR to knock out the genes that blind them to tumors. For instance, deleting the PD-1 gene prevents cancer cells from switching off the immune response.

Problem Statement

Cancer destroys the body through rapid cellular malfunction, localized structural damage, and systemic metabolic collapse. As abnormal cells divide uncontrollably, they form malignant tumors that physically crush vital organs, drain the body’s energy reserves, and hijack the vascular system to spread throughout the body.

Objectives

- The Cas9 Enzyme
- Single Guide RNA (sgRNA)
- The PAM Sequence

Testing and Hypothesis

Clinical Trials: Phase III trials test if a new targeted therapy or immunotherapy statistically significantly extends progression-free survival compared to a placebo or standard of care.

Resulting

It cuts the DNA double strand inside the harmful gene. When the cell attempts to repair this break, it introduces errors that permanently disable the gene.